# ■ InsurCalim AI

Presenter Scripts — slide-by-slide speaking notes

Total run time ≈ 10 minutes · 18 slides · Kasaam Ali · Final-year project review

**How to use:** One box per slide, in order. The blue time-stamp is your running clock — glance at it to stay on pace. Speak at a calm ~140 words a minute; if you're running long, the italic *trim* notes tell you what to cut. The grey line under each script is a delivery cue, not something to read aloud.

SLIDE 1 — 0:00 – 0:35

## Title — InsurCalim AI

Good morning everyone, and thank you for being here. My project is **InsurCalim AI** — a platform that investigates auto-insurance claims using explainable AI. In simple terms: you feed it a claim — the documents, the photos, even video — and instead of a black-box yes-or-no, it hands a human investigator a ranked, fully-cited report that shows its reasoning. It's built around four specialised AI agents, a six-layer fraud check, live market-price checking, and — this is the key part — every conclusion is explained and every final decision stays with a person.

■ *Stand relaxed, make eye contact, don't rush the opening. Point to the four chips as you name them.*

SLIDE 2 — 0:35 – 1:15

## The Problem

So why does this matter? Today, claims handling is slow, costly, and dangerously exposed to fraud. Every claim lands as a messy pile of documents, photos and videos that an adjuster has to read and cross-check *by hand* — millions of times a year. Meanwhile fraud has levelled up: people now submit AI-generated fake photos, fake medical reports, fake valuation certificates that older tools simply were not built to catch. All that manual re-keying becomes a bottleneck that delays honest customers. And when a claim is refused, a black-box score can't be justified to the claimant or the regulator — insurers need an *explanation*, not just a number.

■ *Four cards — gesture across them. Land hard on 'explanation, not just a number'; that's the thesis.*

SLIDE 3 — 1:15 – 1:50

## How Big Is It

And this is not a small problem. Insurance fraud runs to over **eighty billion dollars** a year globally, and by broader estimates over three hundred billion in the US alone — with ninety to a hundred-and-twenty billion of that from property-and-casualty lines like auto. Roughly one in ten P&C loss dollars involves fraud, and a growing quarter to a third of manipulated claims now use AI-faked media. For the ordinary honest customer, that's about a **nine-hundred-dollar-a-year hidden tax** baked into their premiums. So this is a genuine, multi-hundred-billion-dollar pain point.

■ *Let the big numbers breathe — pause after each. Sources are cited at the bottom if asked.*

SLIDE 4 1:50 – 2:20

## Market Opportunity

It's also a fast-moving market. AI in insurance is somewhere around ten to twenty billion dollars today and is forecast to grow to anywhere from eighty-eight up to two-hundred-and-forty-six billion by the early 2030s — a compound growth rate north of thirty percent a year. AI is already shown to cut fraud losses by up to thirty percent. And claims-processing AI specifically is one of the fastest-growing slices inside that. So InsurCalim AI is aimed squarely at a large and accelerating arena.

■ *Keep this one brisk — it's supporting evidence, not the star. Move at a steady pace.*

SLIDE 5 2:20 – 2:55

## Validation — Proven by the Leaders

And I'm not inventing the approach — the biggest players are already proving it works. EY hit seventy-percent auto-extraction with document intelligence. PwC got twenty-nine-percent efficiency gains from explainable damage assessment. Aviva, with McKinsey, ran eighty-plus AI models with a human in the loop and saved sixty million pounds in a single year. Shift Technology, AWS with EXL, Lemonade with three-second settlements — same playbook. My project applies exactly those proven patterns: document intelligence, explainability, and human-in-the-loop — combined into one open pipeline.

■ *Don't read every row — pick two or three, then say 'same playbook.' Table is there for credibility.*

SLIDE 6 2:55 – 3:35

## Our Solution

So here's what InsurCalim AI actually is. It turns a raw claim into a defensible report through four specialised agents wired together as a LangGraph state machine — intake, then timeline, then fraud, then report. It accepts almost anything: text, PDF, Word, images and video, with vision-OCR for scanned documents. At its heart is a six-layer fraud detector. A coverage gate first checks the policy actually covers this and that every required document is present. An amount agent then judges whether the demanded cost is even fair — which matters, because plenty of claims aren't outright fraud, just inflated. And the output is an explainable, cited report that flags and ranks but *never* auto-denies.

■ *This is the core slide — slow down. Six cards; group them as 'agents, intake, fraud, gate, amount, report.'*

SLIDE 7 3:35 – 4:15

## How It Works

Let me walk the pipeline left to right. A claim comes in — documents, photos, video. Agent one, Intake, pulls it all into structured JSON. Agent two, Timeline, orders the events and flags gaps or impossible sequences — like damage dated before the accident. Agent three is the six-layer fraud check. Agent four writes the ranked, cited report. And then it hands off to a human investigator who makes the call. The clever part underneath is a single shared object we call *ClaimState* that flows through the whole graph — so every agent builds on the last, and the human always makes the final decision.

■ *Trace the arrows physically with your hand as you speak each step. End on 'human makes the final decision.'*

SLIDE 8 4:15 – 4:55

## Fraud Detection — Six Layers

Let's open up that fraud agent, because it's the technical heart. Instead of trusting one model, it stacks six independent layers into one hybrid score. Deterministic rules catch hard signals like back-dated or lapsed policies. Perceptual photo-hashing catches recycled or reused damage photos. A machine-learning score — DistilBERT plus a Random Forest trained on thirty-thousand real records — gives a data-driven read. A RAG layer retrieves past known-fraud claims that look like this one. A narrative layer reads the whole file for contradictions across documents. And a deepfake layer flags AI-generated fake images and papers. Six weak signals combined make one strong, defensible judgement.

■ *Count the six on your fingers if it helps. Emphasise 'trained on thirty-thousand real records.'*

SLIDE 9 4:55 – 5:35

## Amount (Quantum) Assessment

Now here's a piece most fraud tools miss entirely. Not every claim is fraud — but a huge number are simply *inflated*. So this agent asks a different question: is the amount being demanded actually fair? It searches *live market prices* on the web for the damaged parts and repairs, builds a line-item fair estimate, and returns a verdict — reasonable, overstated, or needs-more-info. Take the example on screen: the claimant demanded eight-hundred-and-fifty-thousand rupees. The agent priced the real parts — bumper, paint, and so on — and landed near *ninety-five thousand*. Same claim, flagged as overstated, with the pricing shown. That's money saved that a pure fraud filter would never have caught.

■ *The 850k-vs-95k contrast is your best 'wow' moment — pause on it. Let the number sink in.*

SLIDE 10 5:35 – 6:05

## Explainability & Human-in-the-Loop

None of this is a black box — and that's deliberate. The risk score breaks down into named, weighted reason codes, so you see exactly *why* a claim scored the way it did. Every flag carries its evidence and the policy clause it relies on. And the system never auto-denies — it flags and ranks; a human investigator always makes the final call. There's even a built-in AI chat so the investigator can ask grounded, per-claim follow-up questions. Trust is designed in, not bolted on.

■ *This is the ethical backbone of the project — say it with conviction. 'Trust is designed in, not bolted on.'*

SLIDE 11 6:05 – 6:35

## Architecture

Quickly on the architecture, because it's cleaner than it looks. One shared state, one agent graph, and a few supporting stores around it. A resilient multi-provider LLM layer powers the extraction and reasoning. PostgreSQL holds the structured claim data, Chroma powers the RAG retrieval, and the machine-learning models sit alongside. Everything is served through a single login-gated investigator dashboard. So it's modular — each part can be swapped or scaled independently.

■ *Don't read the diagram box by box — give the shape. If time is short, this is a slide you can trim.*

SLIDE 12 6:35 – 7:05

## Technology Stack

On the stack — it's modern and production-shaped. LangGraph and LangChain for orchestration. A multi-provider LLM chain — Z.AI, Gemini, Groq, Claude, OpenRouter — with automatic fallback and key rotation, so no single vendor outage stops the pipeline. Chroma for RAG, PostgreSQL for data, DistilBERT and Random Forest for the ML, Streamlit and a React landing for the front end, and Docker plus AWS for infrastructure. The nice consequence of that fallback design: it runs at near-zero cost on free tiers today, but can scale straight up to paid, higher-accuracy models.

■ *Name-drop confidently but don't linger — the payoff line is 'near-zero cost, scales up.'*

SLIDE 13 7:05 – 7:35

## Live Demo — Product

Let me show you the product itself. This is the front door — a polished React landing page with a single sign-in that takes you into the investigator dashboard. And here's that dashboard: a portfolio overview where an investigator sees their whole book of claims at a glance — volumes, risk levels, and what needs attention. It's not a prototype mock-up; this is the running system.

■ *If doing a LIVE demo, switch to the app now. If not, walk the two screenshots. Have a backup ready.*

SLIDE 14 7:35 – 7:55

## Live Demo — Claims Queue

And this is where the value lands day-to-day: the triaged claims queue. Every claim is scored, ranked, and ordered by risk, so an investigator spends their time on the claims that actually need a human — highest-risk first — instead of working through a random pile. That's the whole promise in one screen: ranked, triaged, explainable.

■ *Short and punchy. Point at the top of the queue — 'highest-risk first.'*

SLIDE 15 7:55 – 8:30

## Cost & ROI

Now the economics, which are honestly the easiest part to defend. Analysing a claim costs *cents*. On free tiers it's effectively zero; on the recommended hybrid tier — Gemini Flash plus Claude Sonnet — it's about seven cents a claim, or seventy dollars per thousand claims, and that tier catches contradictions strongly. Set that against the payoff: catching a *single* inflated or fraudulent claim saves thousands. That's an order-of-magnitude return — cents in, thousands saved.

■ *Land the contrast: 'cents in, thousands saved.' This is your closing-the-sale slide.*

SLIDE 16 8:30 – 9:00

## What’s Built

So where does the project actually stand? The MVP is complete and runs end to end — today. The full pipeline works: multi-format intake, six-layer fraud, amount pricing, the explainable report, and the AI chat. It sits on real data foundations — PostgreSQL and Chroma, with the ML model trained on thirty-thousand real records, using only synthetic and public data, no real personal information. There’s a polished React front door into a secure dashboard, and it all runs at near-zero cost. This is working software, not slides.

■ *Say ‘working software, not slides’ with a small smile — it’s confident, not arrogant.*

SLIDE 17 9:00 – 9:30

## Roadmap & The Ask

Looking ahead — the path from working demo to a real insurer pilot is clear. First, move the primary path onto a paid, higher-accuracy model for production reliability. Second, deploy to the cloud on AWS so a partner can pilot on real infrastructure. Third, and most important, run a pilot on a partner’s permissioned historical claims to validate the fraud-catch rate on real data. And fourth, harden it — audit logging, output-consistency controls, and role-based access. The ask is simple: a data partner and the runway to run that pilot.

■ *State the four steps crisply, then land the ask. Look up on ‘a data partner and the runway.’*

SLIDE 18 9:30 – 10:00

## Conclusion

To wrap up: InsurCalim AI is explainable claim investigation, built to attack a genuine billion-dollar problem. Four agents, a six-layer fraud check, fair-cost pricing, and every decision cited and kept in human hands. It’s built responsibly — synthetic and public data only, and it’s decision *support* for a human investigator, never an automated denial. Thank you very much for your time — I’d be glad to take your questions.

■ *Slow down, smile, thank them genuinely. Then stop talking and invite questions — don’t fill silence.*